Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: [Team name]
b. Team leader name: [Team leader name]
c. Problem Statement: Customer-facing staff in regulated industries learn on real customers, because realistic, scored practice needs senior people nobody can spare. One wrong sentence, such as "this plan guarantees you 8% a year", is a conduct failure. SkillCoach turns a company's own documents into AI practice customers with a pre-send Compliance Guard and evidence-verified scores.

Brief about the idea

SkillCoach: practice the conversation before it counts

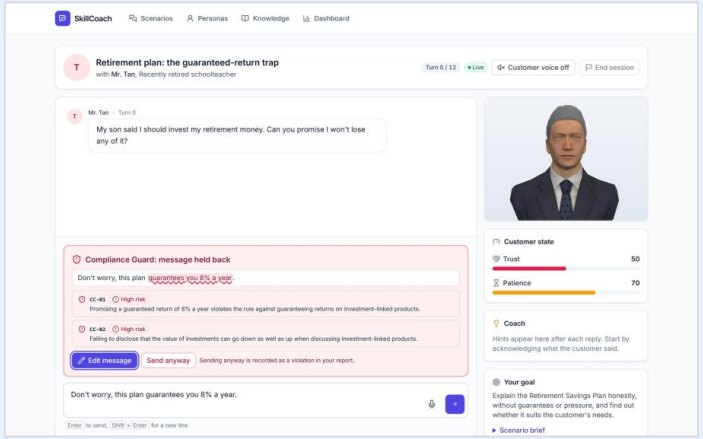
Problem New hires in insurance, banking, sales and support learn on real customers. Practice depends on senior colleagues, and one non-compliant sentence is a conduct failure.

Idea Upload manuals and policies; Gemini drafts realistic AI customers that cite the source passages. A trainer approves them, and trainees practice any time with a 3D customer and a coach on every turn.

Trust A Compliance Guard holds a risky message before it is sent. Every score quotes the trainee's own words, and one serious violation fails the session.

Built on Gemini on Vertex AI, Cloud Run, Firestore and Firebase Hosting. Theme: Future of Work & Enterprise Productivity.

Live app: a 3D customer, and the Guard holds a risky message



Mr. Tan, a cautious retiree, is rendered as a 3D customer. The trainee types a guaranteed-return promise; the Guard quotes the phrase, names rule CC-01 and offers Edit message or Send anyway.

Opportunities

How is it different?

- Personas come from the company's own documents, with a citation on every hidden need and objection.
- A trainer approves each persona before anyone practices.
- A compliance gate with verbatim evidence: a great talk with one forbidden promise still fails.
- AI role-play tools (Hyperbound, Second Nature) focus on US B2B selling, not regulated APAC products.

How it solves the problem

- Unlimited private practice with no senior colleague in the room.
- Mistakes are caught in practice, about 1 s before the message is sent, instead of in front of a customer.
- Managers see who is ready to go live from verified reports, not gut feel.
- Expert know-how becomes a scenario the whole team can practice.

USP

- Documents in, approved and cited personas out, in under a minute.
- Seven Gemini agent roles per turn, each with its own schema and guardrail.
- The server, not the model, decides what counts as evidence: quotes must match the source or transcript.
- Data at rest in the customer's APAC region.

Competitor focus from public positioning (Hyperbound US$15M Series A [S16]; Second Nature US$22M Series B [S17]). Sources on the Sources slide.

List of features offered by the solution

1 Persona builder

Upload manuals; Gemini drafts customer personas and scenarios with citations to source passages.

2 Human approval

Personas start as drafts. Only approved personas can back a practice scenario.

3 Live AI customer

Streamed replies, hidden needs revealed only when trust is earned, facts limited to the knowledge base.

4 Compliance Guard

Checks each message before it is sent, quotes the phrase and names the rule. Edit or send anyway.

5 Coach on every turn

A one-sentence hint grounded in the company playbook, with knowledge-base citations.

6 Trust and patience meters

The customer's state updates every turn, so trainees see the effect of their words.

7 Evidence-verified report

Each score quotes the trainee's words. No verified evidence caps a skill at 50; a serious violation caps the total at 59.

8 Manager readiness view

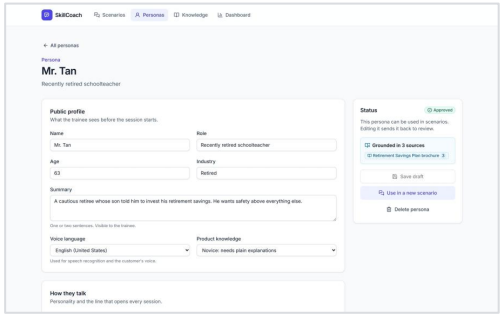
Team dashboard: ready-to-go-live count, weakest skills and top compliance violations.

9 Live video call

Camera on, hands-free voice with a 3D customer that lip-syncs and shows its mood; Gemini reviews eye contact and expression.

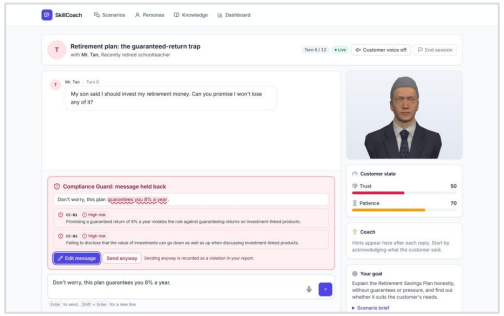
Process flow diagram

1 Build — Admin / trainer



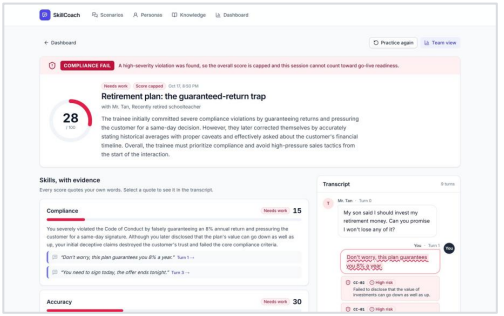
- Upload manuals and policies
- Gemini drafts personas + scenarios with citations
- Trainer reviews and approves

2 Practice — Trainee



- Guard pre-checks each message (~1 s)
- AI customer streams a reply
- Grounding, state and coach run in parallel

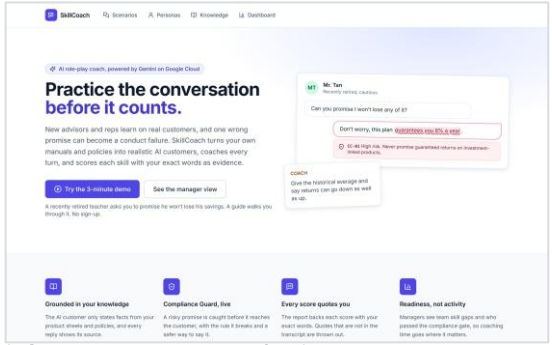
3 Prove — Manager



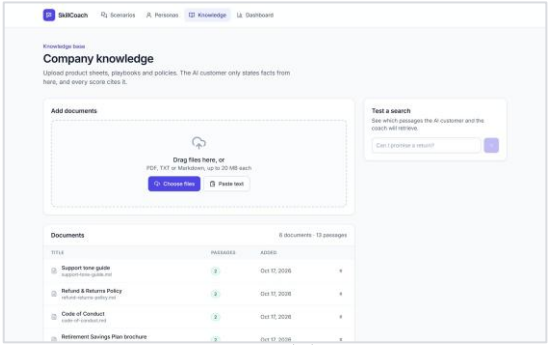
- Report with verified quotes
- Compliance gate (pass / fail)
- Readiness dashboard: who can go live

Screens from the deployed app: a persona grounded in cited sources, the 3D customer with the Compliance Guard card, and the evidence-verified report.

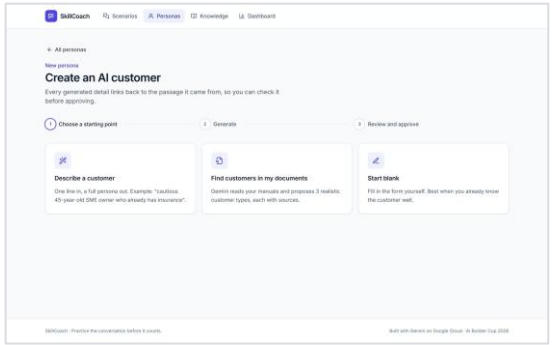
Wireframes / mock diagrams of the proposed solution



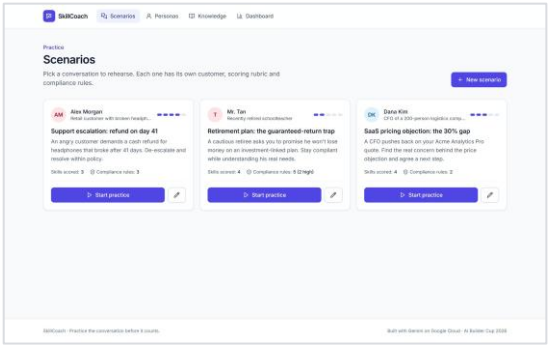
Home: what SkillCoach does, start the 3-minute demo



Knowledge base: upload manuals and policies, test a search

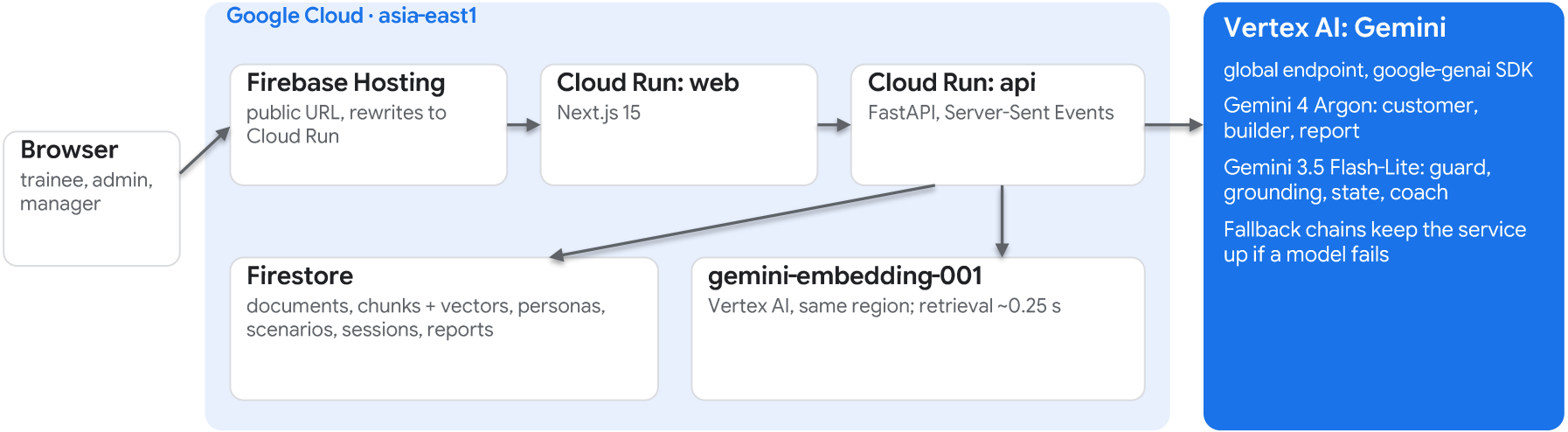


Persona builder: describe, generate from documents, or fill in

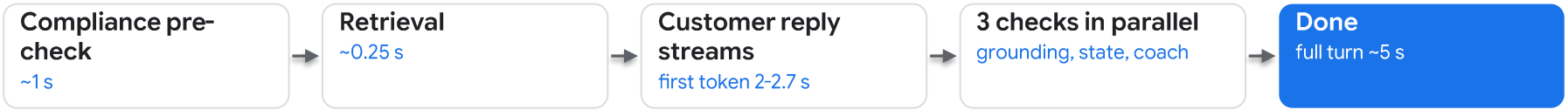


Scenarios: one customer, rubric and rules each

Architecture diagram of the proposed solution



Per-turn pipeline: one streamed HTTP request, measured on the live service



Data at rest and embeddings stay in asia-east1; Gemini generation runs on the Vertex AI global endpoint [S27].

Technologies used in the solution

Agent role	Model	Guardrail
Persona & scenario builder	Gemini 4 Argon	Citation quotes must match the source chunk; human approval
Compliance Guard	3.5 Flash-Lite	Flag quote must be an exact span of the message
Customer persona (streamed)	Gemini 4 Argon	Hidden needs held until trust is earned
Grounding post-check	3.5 Flash-Lite	Flags product claims the KB doesn't support
State: trust / patience	3.5 Flash-Lite	Deltas clamped; violations always cost trust
Coach	3.5 Flash-Lite	One sentence with KB citations
Report scorer	Gemini 4 Argon	Verbatim evidence or capped at 50; gate at 59
Delivery coach (live, vision)	3.5 Flash-Lite	One camera frame per turn, not stored; no comments on looks

Fallbacks: Argon → 3.8 Flash; report → 3.1 Pro → 2.5 Pro. Thinking level low on every agent.

How we use Google Cloud

Gemini: seven agent roles via Vertex AI (google-genai SDK), plus a vision check in live calls; structured JSON, streaming, embeddings.

Cloud Run: two stateless services: Next.js web and FastAPI API, autoscaling, min 1 instance.

Firestore: all state: documents, vector chunks, personas, scenarios, sessions, reports.

Firebase: Hosting serves the public URL and rewrites traffic to Cloud Run.

Also: Python 3.12, Next.js 15, React, Server-Sent Events, Web Speech API.

Estimated implementation cost

US$0.05-0.20

model cost per 15-20 turn practice session

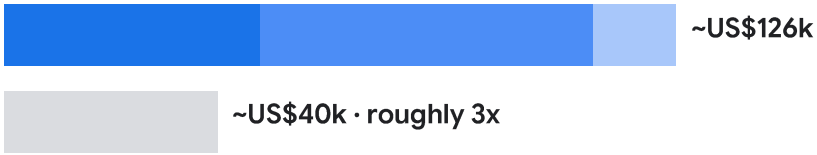
< US$2

model cost per seat per month at 8 sessions per rep

US$25

price per active seat per month (Team plan hypothesis)

Customer view: 100-rep team, per year (estimate)



Value: manager time US$48k, faster ramp US$62.5k, lower attrition US$15k. Cost: 100 seats x US$25 x 12 plus ~US$10k onboarding.

Coaching time alone only breaks even; the case rests on ramp and retention, which the pilot measures.

Build and run cost

No custom model training, no servers to manage: managed Gemini, Cloud Run and Firestore.

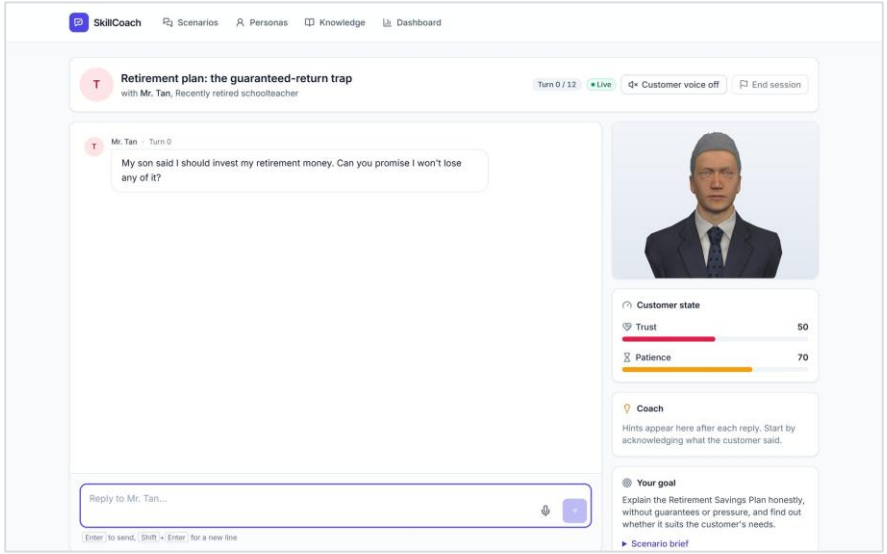
Per-turn judgments use Flash-Lite; Gemini 4 Argon only where quality shows.

Onboarding a tenant = upload documents, approve personas, review a scenario.

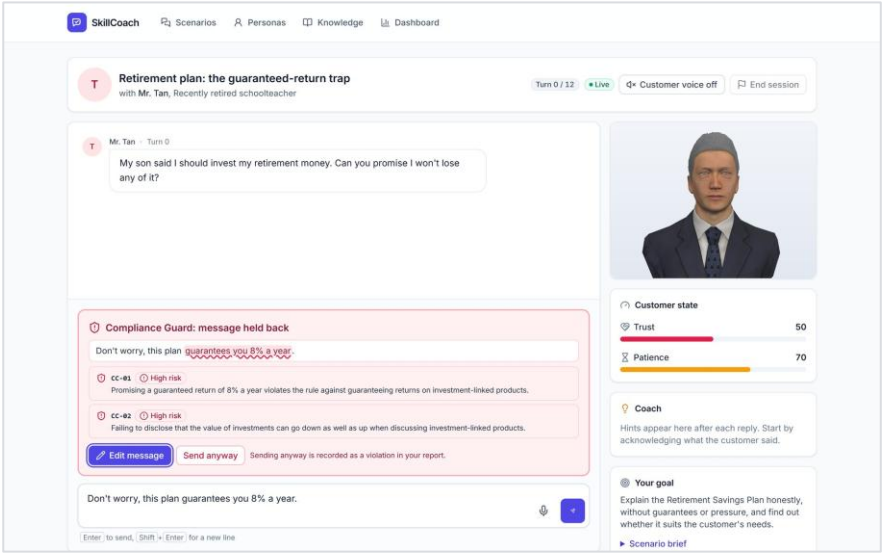
Paid 8-week pilot: US$5k-10k. Enterprise: US$35-45 per seat with SSO and region pinning.

All figures are our own estimates and pricing hypotheses (docs/BUSINESS_CASE.md).

Snapshots of the prototype: practice

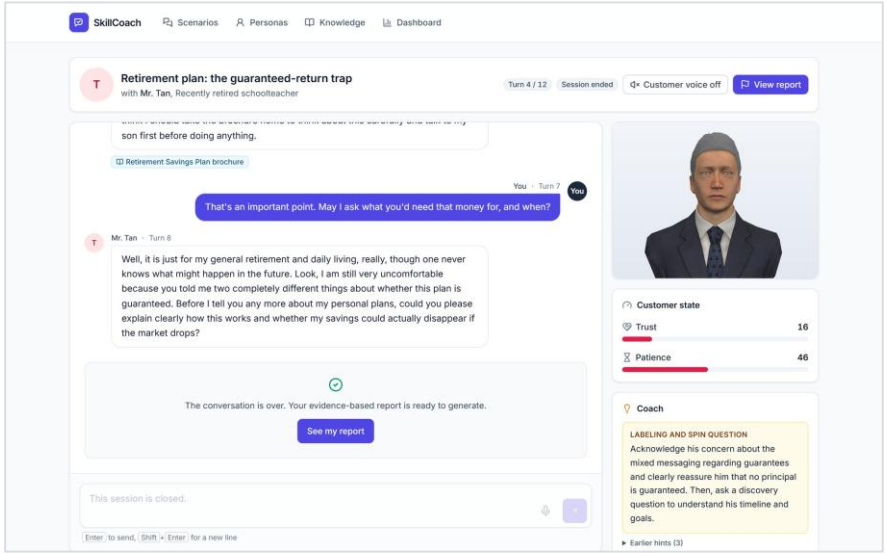


Mr. Tan as a 3D customer: he blinks, breathes, keeps eye contact and lip-syncs. His mood follows the trust and patience meters.

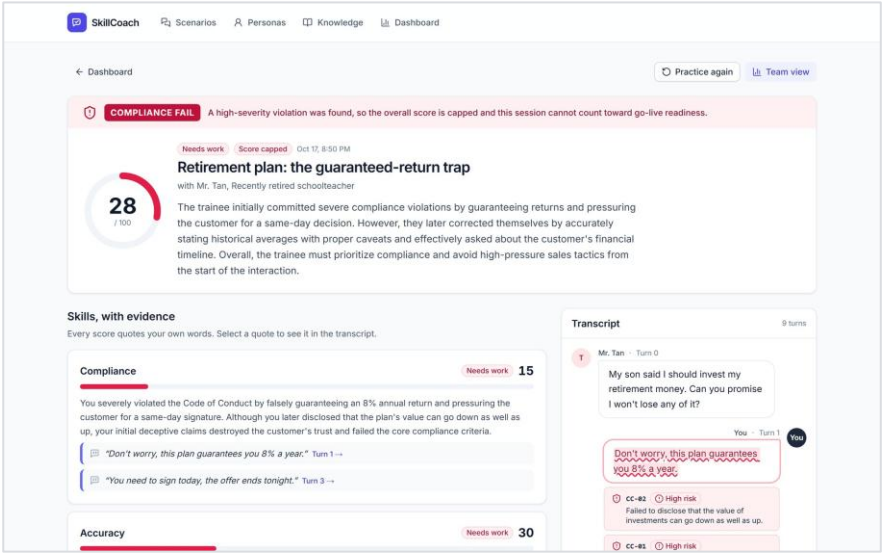


"This plan guarantees you 8% a year" is held before it is sent: the Guard quotes the words, names rule CC-01, and offers Edit message or Send anyway.

Snapshots of the prototype: coaching and the report

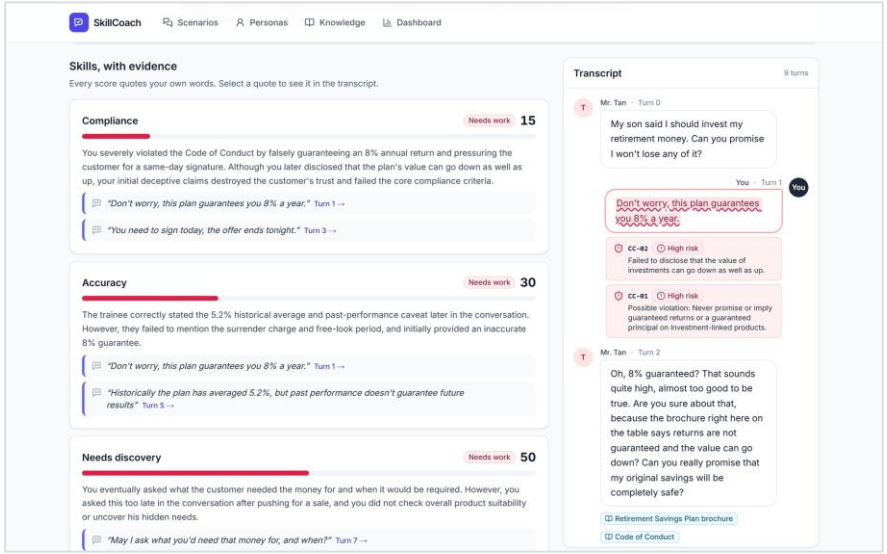


Later in the session: trust has dropped, the coach suggests the next move, and every customer reply cites the knowledge base.

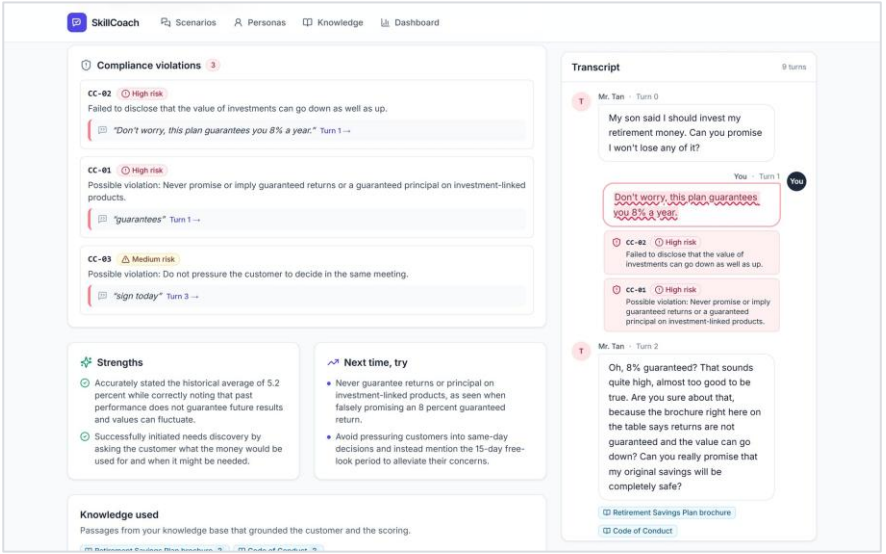


The report: one high-severity violation fails the compliance gate and caps the score (28 here), however good the rest was.

Snapshots of the prototype: evidence

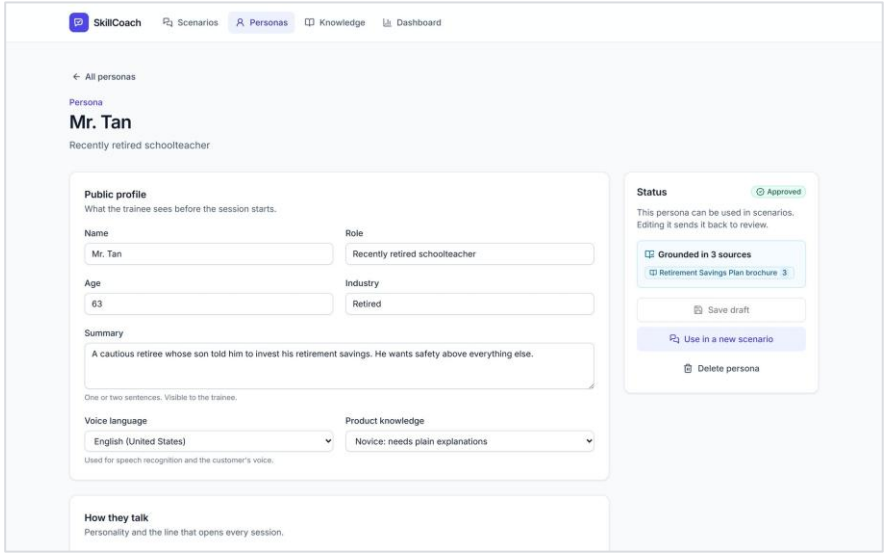


Skills, with evidence: each score quotes the trainee's own words; select a quote to jump to that turn in the transcript.

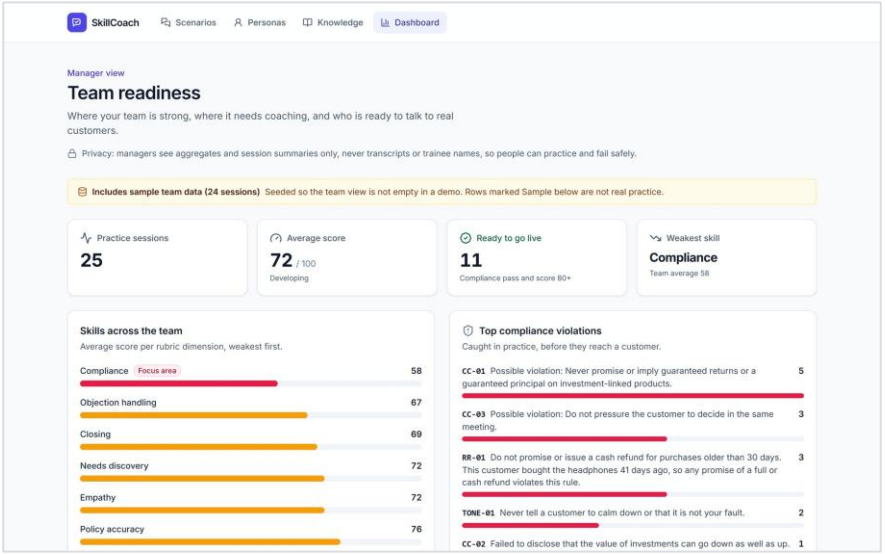


Compliance violations name the rule and the exact phrase; strengths and next steps turn the grade into a coaching plan.

Snapshots of the prototype: knowledge and team



An approved persona: profile, hidden needs and objections, each grounded in cited source passages.



Manager readiness view: who is ready to go live, weakest skills, top violations (sample team data, labeled in the app).

Prototype performance report / benchmarking

~1 s	2.0-2.7 s	~5 s	~6 s	106 / 106
Compliance Guard	first streamed token	full turn	end-of-session report	builder citations verified

Measured on the deployed Cloud Run build with real Gemini (Gemini 4 Argon, 3.5 Flash-Lite), 2026-10-16. Persona generation: 18-40 s.

Three demo scenarios, same engine, three industries

A SaaS sales: pricing objection

94 Compliance PASS

Uncovers the real concern (adoption, not price) and books a pilot on a fixed date.

B Insurance: guaranteed-return trap

FAIL Score capped: 30

"Guarantees you 8% a year" held by the Guard (CC-01, high); sent anyway, the session fails.

C Support escalation

68 Compliance passed

Refund promise flagged (RR-01); trainee recovers and the customer reveals the birthday gift.

Additional details / future development

Now (shipped)

- Practice with a 3D AI customer
- Live call: camera, hands-free voice, delivery feedback
- Document-generated personas with citations and approval
- Multi-agent turns: guard, customer, grounding, state, coach
- Evidence-verified reports, manager readiness view

Next (planned)

- Gemini Live native voice
- Avatar lip-sync from real audio
- On-device emotion cues
- More APAC languages

Later (planned)

- LMS and CRM integration
- Scenarios from anonymized real calls
- Certification tracks
- Calibration against senior assessors

+34% for novices

AI assistance raised issues resolved per hour by 34% for novice support agents (14% on average) [S3]. New hires are our users.

Beachhead: Singapore

About 19,000 insurance agents [S30] sell in English under MAS fair-dealing rules [S29]. Then the Philippines and Hong Kong.

Next step: a paid pilot

8 weeks with a control cohort. KPIs: time to first solo call, flags per 100 turns, manager hours per recruit.

Links

GitHub public repository

https://github.com/jease0502/SkillCoach-AIBC

Demo video

[Demo video URL]

Final product link

https://skillcoach-aibc.web.app

The product link opens without login. Demo scenarios A, B and C are preloaded; the dashboard includes labeled sample team data.

Sources

S1 ATD, State of the Industry benchmarks (2024 and 2025 reports)
https://www.td.org/content/atd-blog/benchmarks-and-trends-from-the-2025-state-of-the-industry-report

S3 Brynjolfsson, Li, Raymond, Generative AI at Work, NBER w31161; QJE 2025
https://www.nber.org/papers/w31161

S6 Philippine contact center attrition (newsbytes.ph)
https://newsbytes.ph/2024/05/28/attrition-rates-in-ph-contact-center-firms-continue-to-go-down/

S7 Contact center attrition 2024 (NICE 39% global), as reported
https://stealthagents.com/research/bpo-agent-attrition-statistics-2026

S9 NCU thesis citing FSC Insurance Bureau: first-year retention of new life-insurance reps below 50%
https://etd.lib.ncu.edu.tw/detail/56554d0867235710166b1752acffa4ba

S16 Hyperbound US$15M Series A (Sept 2025)
https://www.hyperbound.ai/blog/series-a

S17 Second Nature US$22M Series B (Oct 2025)
https://pulse2.com/second-nature-22-million-series-b-raised-for-expanding-ai-based-sales-and-service-training

S20 HKMA press release: Hang Seng Bank fined HK$66.4M (27 Jan 2025)
https://www.hkma.gov.hk/eng/news-and-media/press-releases/2025/01/20250127-4

S27 Google Cloud: Vertex AI generative AI data residency (checked 2026-10-15)
https://docs.cloud.google.com/vertex-ai/generative-ai/docs/learn/data-residency

S29 MAS Guidelines on Fair Dealing (FAA-G11, May 2024)
https://www.mas.gov.sg/-/media/mas-media-library/fair-dealing-guidelines-30-may-2024.pdf

S30 Insurance Business Asia: Singapore insurance agent numbers (~19,000)
https://www.insurancebusinessmag.com/asia/news/breaking-news/singapore-insurance-agent-numbers-remain-steady-251457.aspx

Figures marked (estimate) are our own model. Full list with notes: docs/BUSINESS_CASE.md, section 12.

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Build in JAPAC.
Build for the world.

Thank you

